

Highlights

Subseasonal Temperature Forecasting in Andean Stations: A Benchmark of Machine Learning and Deep Learning Models with Anomaly Decomposition

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- Zero-leakage climatology isolates thermal anomaly signal
- Sliding 2025 windows benchmark ensembles vs LSTM vs TFT
- Public IGP Huayao hourly data enables full replication
- Ablation quantifies physical feature group contributions
- Probabilistic skill supports subseasonal planning use

Subseasonal Temperature Forecasting in Andean Stations: A Benchmark of Machine Learning and Deep Learning Models with Anomaly Decomposition

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ABSTRACT

TODO (draft abstract): Subseasonal temperature forecasting supports energy planning and agriculture in the Andes, where steep terrain produces strong local gradients between stations. We benchmark classical machine learning and deep learning models for one-to-four-week horizons on a shared protocol over an Andean station network (2018–2025), with and without anomaly decomposition as a preprocessing stage. Every reported number is anchored to a machine-readable artefact under `experiments/`.

1. Introduction

Short- and subseasonal-horizon temperature forecasts drive energy demand planning, frost alerts and agricultural scheduling in the Andes, where elevation and aspect create large temperature differences between nearby stations. Data-driven methods have become the default baseline for such tasks: tree-based models for tabular regression and recurrent or temporal convolutional architectures for sequence learning (Reichstein, Camps-Valls, Stevens, Jungstein, Denzler and Mahecha, 2019; Salinas, Flunkert, Gasthaus and Januschowski, 2020).

This paper contributes a reproducible benchmark of machine learning and deep learning models for temperature forecasting on an Andean station network, with a controlled ablation of anomaly decomposition as a preprocessing stage. All results in this manuscript are produced by the pipeline under `experiments/` and are listed in `manifest.yaml` as auditable claims.

2. Data

The benchmark uses hourly records from Andean meteorological stations (2018–2025), distributed as `data/raw/dataset.csv`. TODO: document the stations, gaps, quality control and train/validation/test split windows.

Table 1

Stations included in the benchmark. TODO: fill from `data/raw/dataset.csv` and keep the table in sync with the pipeline.

Station	Altitude (m.a.s.l.)	Period	Records	Missing
TODO-station-1	0000	2018–2025	00000	00.0%
TODO-station-2	0000	2018–2025	00000	00.0%
TODO-station-3	0000	2018–2025	00000	00.0%

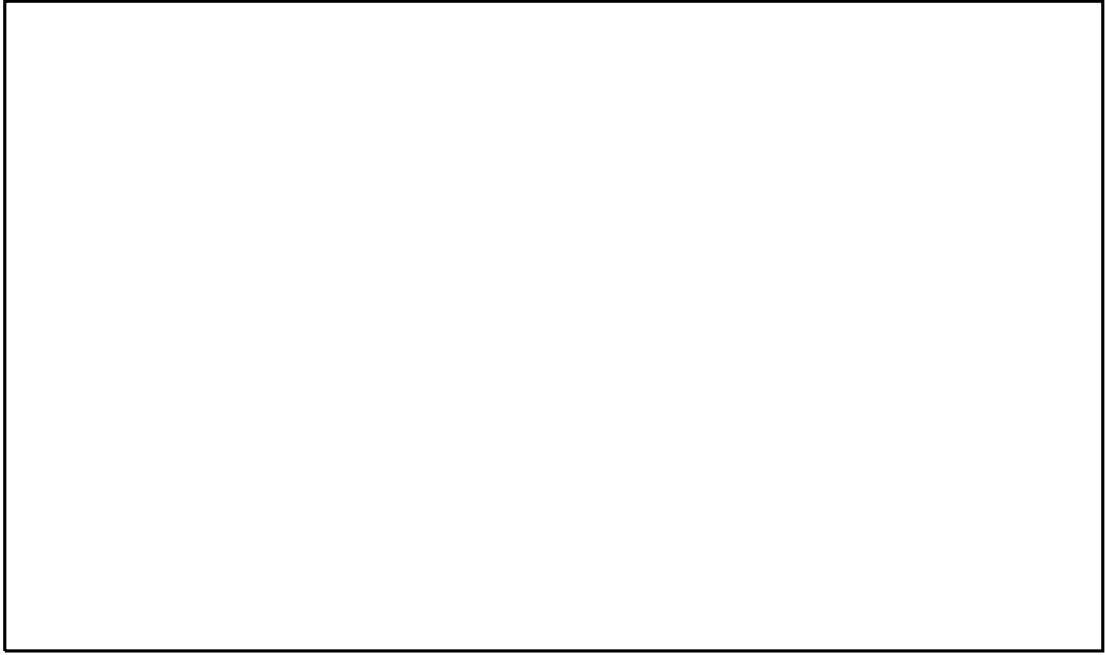


Figure 1: TODO: replace with the anomaly-decomposition and training workflow diagram (source: experiments/).

3. Methodology

1 summarizes the pipeline and 1 lists the stations. Each forecasting model is trained per horizon (1, 2, 3 and 4 weeks) under the same split protocol, once on the raw series and once on the residuals after anomaly decomposition. TODO: name the concrete models, hyperparameter search budget and evaluation metrics (MASE, MAE, RMSE).

4. Results

TODO: report the benchmark table and learning curves; every number must trace back to a file under outputs/ (see claims in manifest.yaml, e.g. claim-1).

5. Discussion

TODO: where decomposition helps or hurts per model family and horizon, and how the stations' altitude modulates the ranking.

6. Conclusions

TODO: headline findings and practical guidance for station-network forecasting.

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Data availability

TODO: repository or DOI for `data/raw/dataset.csv` and the exact command that regenerates every figure and table of this paper.

References

- Reichstein, M., Camps-Valls, G., Stevens, B., Jungstein, M., Denzler, J., Mahecha, M., 2019. Deep learning and process understanding for data-driven earth system science. *Nature* 566, 192–194. doi:10.1038/s41586-019-0912-1.
- Salinas, D., Flunkert, V., Gasthaus, J., Januschowski, T., 2020. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting* 36, 1181–1191. doi:10.1016/j.ijforecast.2019.07.001.